

2483. Minimum Penalty for a Shop

Difficulty: Medium

Link: <https://leetcode.com/problems/minimum-penalty-for-a-shop/?envType=daily-question&envId=2025-12-26>

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Problem:

You are given the customer visit log of a shop represented by a **0-indexed** string `customers` consisting only of characters `'N'` and `'Y'`:

- if the i^{th} character is `'Y'`, it means that customers come at the i^{th} hour
- whereas `'N'` indicates that no customers come at the i^{th} hour.

If the shop closes at the j^{th} hour ($0 \leq j \leq n$), the **penalty** is calculated as follows:

- For every hour when the shop is open and no customers come, the penalty increases by 1.
- For every hour when the shop is closed and customers come, the penalty increases by 1.

Return the **earliest** hour at which the shop must be closed to incur a **minimum** penalty.

Note that if a shop closes at the j^{th} hour, it means the shop is closed at the hour j .

Example 1:

Input: customers = "YNYN"

Output: 2

Explanation:

- Closing the shop at the 0th hour incurs in $1+1+0+1 = 3$ penalty.
- Closing the shop at the 1st hour incurs in $0+1+0+1 = 2$ penalty.
- Closing the shop at the 2nd hour incurs in $0+0+0+1 = 1$ penalty.
- Closing the shop at the 3rd hour incurs in $0+0+1+1 = 2$ penalty.
- Closing the shop at the 4th hour incurs in $0+0+1+0 = 1$ penalty.

Closing the shop at 2nd or 4th hour gives a minimum penalty. Since 2 is earlier, the optimal closing time is 2.

Example 2:

Input: customers = "NNNNN"

Output: 0

Explanation: It is best to close the shop at the 0th hour as no customers arrive.

Example 3:

Input: customers = "YYYYY"

Output: 4

Explanation: It is best to close the shop at the 4th hour as customers arrive at each hour.

Constraints:

- $1 \leq \text{customers.length} \leq 10^5$
- customers consists only of characters 'Y' and 'N'.